NAME: VINAY BHARGAV REDDY ID NO:2400031280 SEC NO : 12

FULL STACK DEVLOPMENT

Skill-6: Spring MVC Web Request Handling Demo – Online Library System

Purpose

Demonstrate Spring Boot MVC request handling using @RestController, @GetMapping, @PostMapping, @PathVariable, and @RequestParam.
Show how to build REST endpoints for an online library system, handling book data and user requests.

Tech stack

Java 21, Maven 3.x, Spring Boot 3.2.4, spring-boot-starter-web

---

Project files (full contents)

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
                    https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>skill-6-spring-mvc-demo</artifactId>
  <version>1.0.0</version>
  <packaging>jar</packaging>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.2.4</version>
    <relativePath/>
  </parent>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
```

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```
  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
  </build>
</project>
```

Book.java

```
package main.java.com.example;

public class Book {
  private int id;
  private String title;
  private String author;

  public Book() {}

  public Book(int id, String title, String author) {
    this.id = id;
    this.title = title;
    this.author = author;
  }

  public int getId() { return id; }
  public void setId(int id) { this.id = id; }

  public String getTitle() { return title; }
  public void setTitle(String title) { this.title = title; }

  public String getAuthor() { return author; }
  public void setAuthor(String author) { this.author = author; }
```

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}

LibraryController.java

```
package main.java.com.example;

import org.springframework.web.bind.annotation.*;
import java.util.*;

@RestController
public class LibraryController {
    private List<Book> bookList = new ArrayList<>();

    @GetMapping("/welcome")
    public String welcome() {
        return "Welcome to the Online Library System";
    }

    @GetMapping("/count")
    public int count() {
        return bookList.size();
    }

    @GetMapping("/price")
    public double price() {
        return 299.99;
    }

    @GetMapping("/books")
    public List<String> getBooks() {
        List<String> titles = new ArrayList<>();
        for (Book b : bookList) {
            titles.add(b.getTitle());
        }
        return titles;
    }
```

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```
    @GetMapping("/books/{id}")
    public Book getBookById(@PathVariable int id) {
        for (Book b : bookList) {
            if (b.getId() == id) return b;
        }
        return null;
    }

    @GetMapping("/search")
    public String searchBook(@RequestParam String title) {
        return "Search confirmed for book: " + title;
    }

    @GetMapping("/author/{name}")
    public String getAuthor(@PathVariable String name) {
        return "Author: " + name;
    }

    @PostMapping("/addbook")
    public String addBook(@RequestBody Book book) {
        bookList.add(book);
        return "Book added: " + book.getTitle();
    }

    @GetMapping("/viewbooks")
    public List<Book> viewBooks() {
        return bookList;
    }
}
```

Skill6Application.java

```
package main.java.com.example;

import org.springframework.boot.SpringApplication;
```

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```
import org.springframework.boot.autoconfigure.SpringBootApplication;


@SpringBootApplication
public class Skill6Application {
    public static void main(String[] args) {
        SpringApplication.run(Skill6Application.class, args);
    }
}
```

application.properties

```
server.port=8081
```

Build

From skill-6 folder:
mvn clean install

Run (with Maven [INFO] logs) (PowerShell)

From skill-6 folder:
mvn spring-boot:run

Expected console flow

- Spring Boot starts, shows [INFO] logs and port info.
- All endpoints available at http://localhost:8081/
- Test endpoints in browser/Postman:
  - /welcome returns welcome message.
  - /addbook (POST) adds a book.
  - /viewbooks returns all added books.
  - /books/{id} returns book details.
  - /search?title=xyz confirms search.
  - /author/{name} returns author info.

Expected Output

- JSON responses for all endpoints.
- Example:
  - /welcome: "Welcome to the Online Library System"
  - /addbook (POST JSON): {"id":1,"title":"Java","author":"Hareesh"}
  - /viewbooks: [{"id":1,"title":"Java","author":"Hareesh"}]
  - /books/1: {"id":1,"title":"Java","author":"Hareesh"}

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What was done

- Created a Spring Boot Maven project with spring-boot-starter-web dependency.
- Implemented Book model and LibraryController with all required endpoints.
- Used @RestController, @GetMapping, @PostMapping, @PathVariable, @RequestParam.
- Configured port in application.properties to avoid conflicts.
- Built and ran the project, tested endpoints in browser/Postman.

OUTPUTS:

- Terminal screenshot showing [INFO] logs and successful run.
- Postman/browser screenshots showing JSON responses for endpoints

Output screenshots :

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New folder - skill-6-spring-mvc-demo/src/main/java/com/klef/Skill6Application.java - Spring Tools for Eclipse

File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer | Project Explorer

- skill-6-spring-mvc-demo (in skill-6) [boot]
  - src/main/java
    - com.klef
      - Book.java
      - LibraryController.java
      - Skill6Application.java
  - src/main/resources
    - application.properties
  - src/test/java
  - src/test/resources
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations
  - JRE System Library [JavaSE-21]
  - src
    - main
    - test
  - target
  - pom.xml

Boot Dashboard

Type tags, projects, or working set names to match (incl. * and ? wildc

- local

Skill6Application.java | skill-6-spring-mvc-demo/pom.xml | Book.java | LibraryController.java | application.properties

```
1  package com.klef;
2
3  import org.springframework.boot.SpringApplication;
5
6  @SpringBootApplication
7  public class Skill6Application {
8      public static void main(String[] args) {
9          SpringApplication.run(Skill6Application.class, args);
10     }
11 }
12
```

Problems | Javadoc | Declaration | Console

skill-6-spring-mvc-demo - Skill6Application [Spring Boot App] C:\spring-tools-for-eclipse-5.0.1.RELEASE-e4.38.0-win32.win32.x86_64\sts-5.0.1.RELEASE\plugins\org.eclipse.justj.openjdk.hotspo

```
  .   ____          _            __ _ _
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 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::                (v3.2.4)

2026-03-12T22:52:55.391+05:30  INFO 7368 --- [           main] com.klef.Skill6Application               : Starting Skill6Application using
2026-03-12T22:52:55.393+05:30  INFO 7368 --- [           main] com.klef.Skill6Application               : No active profile set, falling b
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.apache.tomcat.jni.Library in an unnamed module (file:/C:/Users/LENOVO/.m2/repositor
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

2026-03-12T22:52:56.070+05:30  INFO 7368 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 808
2026-03-12T22:52:56.087+05:30  INFO 7368 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-03-12T22:52:56.087+05:30  INFO 7368 --- [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache
2026-03-12T22:52:56.125+05:30  INFO 7368 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded Web
2026-03-12T22:52:56.125+05:30  INFO 7368 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: init
2026-03-12T22:52:56.434+05:30  INFO 7368 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port 8083 (htt
2026-03-12T22:52:56.440+05:30  INFO 7368 --- [           main] com.klef.Skill6Application               : Started Skill6Application in 1.4
```
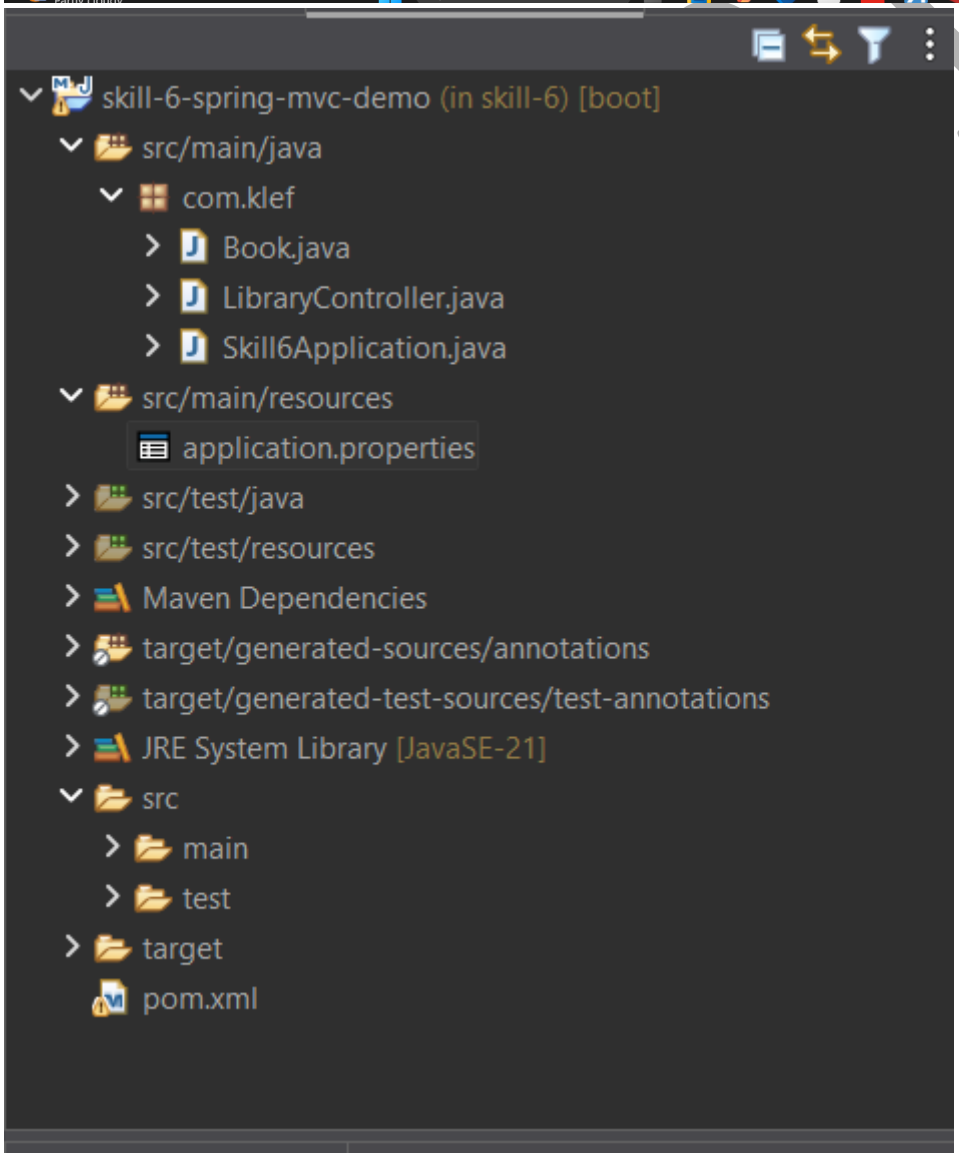

Writable | Smart Insert | 12 : 1 : 311

26°C Partly cloudy | Search | ENG IN | 22:53 12-03-2026



New folder - skill-6-spring-mvc-demo/src/main/java/com/klef/LibraryController.java - Spring Tools for Eclipse

File Edit Source Refactor Navigate Search Project Run Window Help

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      - Book.java
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  - src/main/resources
    - application.properties
  - src/test/java
  - src/test/resources
  - Maven Dependencies
  - target/generated-sources/annotations
  - target/generated-test-sources/test-annotations
  - JRE System Library [JavaSE-21]
  - src
    - main
    - test
  - target
  - pom.xml

Boot Dashboard

- local

Skill6Application.java | skill-6-spring-mvc-demo/pom.xml | Book.java | LibraryController.java | application.properties

```java
package com.klef;

import org.springframework.web.bind.annotation.*;

@RestController
public class LibraryController {
    private List<Book> bookList = new ArrayList<>();

    @GetMapping("/welcome")
    public String welcome() {
        return "Welcome to the Online Library!";
    }

    @GetMapping("/count")
    public int count() {
        return 5; // Sample total books
    }

    @GetMapping("/price")
    public double price() {
        return 299.99; // Sample book price
    }

    @GetMapping("/books")
    public List<String> getBooks() {
        return Arrays.asList("Java Basics", "Spring Boot", "Hibernate", "Microservices", "RESTful APIs");
    }

    @GetMapping("/books/{id}")
```

Problems | Javadoc | Declaration | Console

skill-6-spring-mvc-demo - Skill6Application [Spring Boot App] C:\spring-tools-for-eclipse-5.0.1.RELEASE-e4.38.0-win32.win32.x86_64\sts-5.0.1.RELEASE\plugins\org.eclipse.justj.openjdk.hotspo

```
2026-03-12T22:52:56.070+05:30  INFO 7368 --- [          main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port 808
2026-03-12T22:52:56.087+05:30  INFO 7368 --- [          main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-03-12T22:52:56.087+05:30  INFO 7368 --- [          main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache
2026-03-12T22:52:56.125+05:30  INFO 7368 --- [          main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded Web
2026-03-12T22:52:56.125+05:30  INFO 7368 --- [          main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: init
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2026-03-12T22:52:56.440+05:30  INFO 7368 --- [          main] com.klef.Skill6Application               : Started Skill6Application in 1.4
```

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26°C Partly cloudy | Search | ENG IN | 22:56 12-03-2026

Kishtipati-Vinay-Bhar | localhost:8083/welco | localhost:8082

localhost:8083/welcome

Welcome to the Online Library!

Kishtipati-Vinay- | localhost:8083/w | localhost:8083/c | localhost:8082

localhost:8083/count

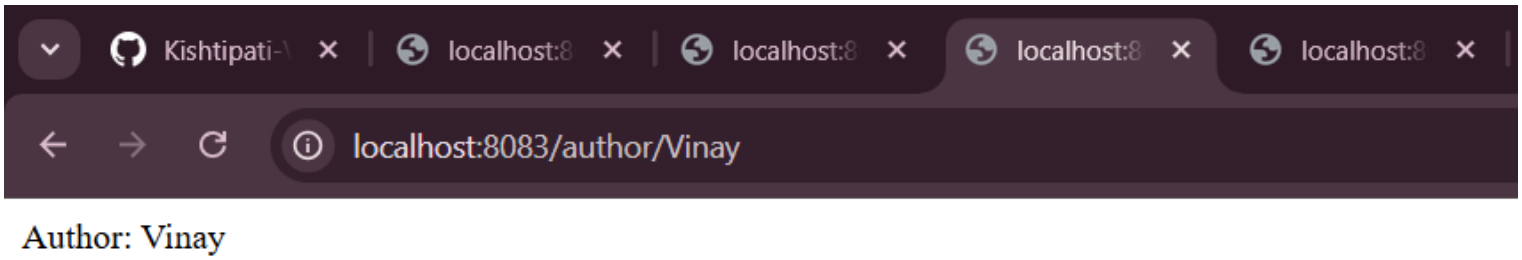
Pretty print

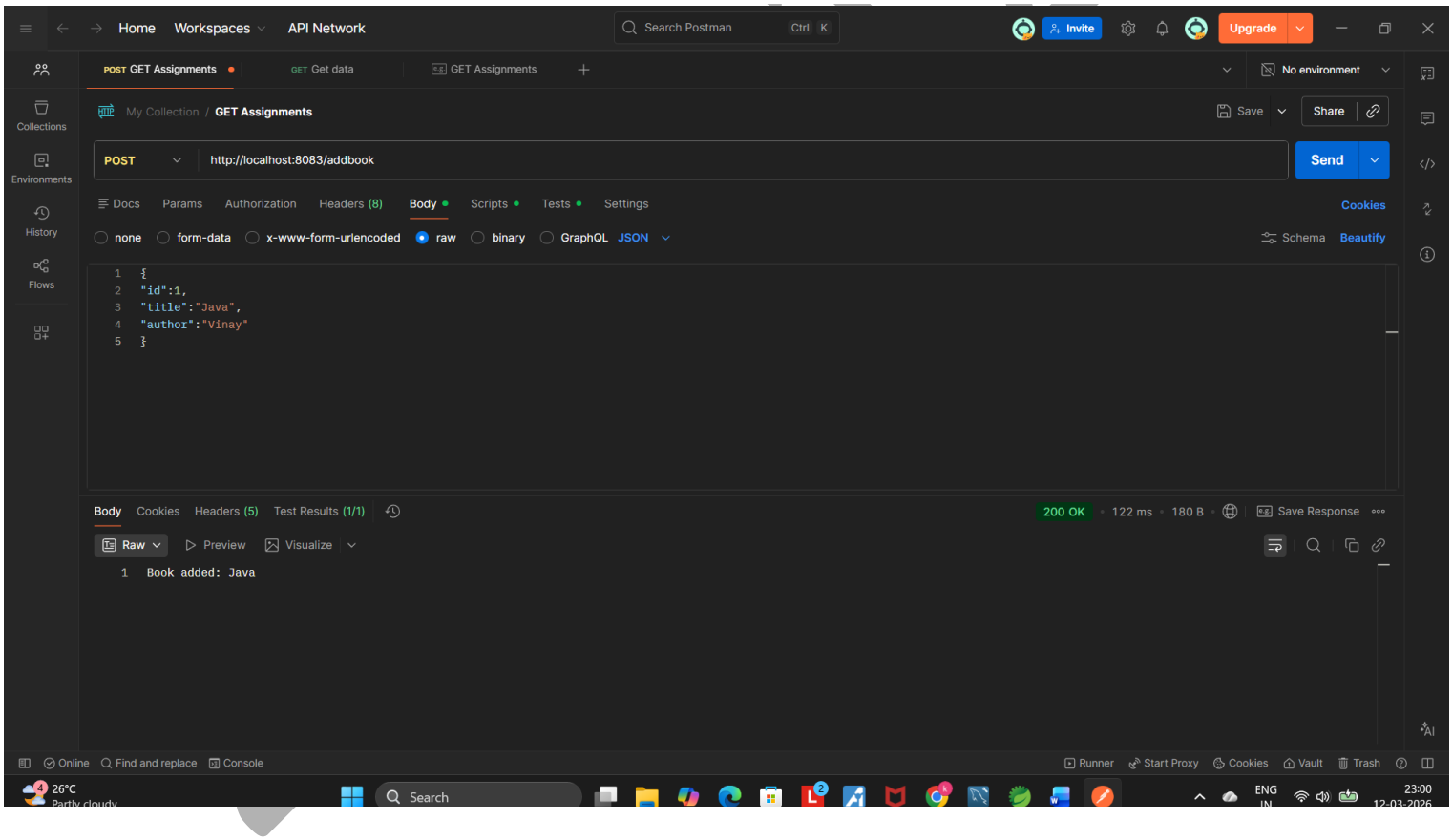
5

Kishtipati-Vin | localhost:808 | localhost:808 | localhost:808 | localhost:808

localhost:8083/search?title=Java

Search confirmed for book: Java


Kishtipati-\
localhost:8
localhost:8
localhost:8
localhost:8
localhost:8083/author/Vinay
Author: Vinay


Home
Workspaces
API Network
Search Postman
Ctrl K
Invite
Upgrade
POST GET Assignments
GET Get data
GET Assignments
No environment
Collections
Environments
History
Flows
My Collection / GET Assignments
Save
Share
POST
http://localhost:8083/addbook
Send
Docs
Params
Authorization
Headers (8)
Body
Scripts
Tests
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
JSON
Schema
Beautify
{
"id":1,
"title":"Java",
"author":"Vinay"
}
Body
Cookies
Headers (5)
Test Results (1/1)
200 OK
122 ms
180 B
Save Response
Raw
Preview
Visualize
Book added: Java
Online
Find and replace
Console
Runner
Start Proxy
Cookies
Vault
Trash
26°C
Partly cloudy
Search
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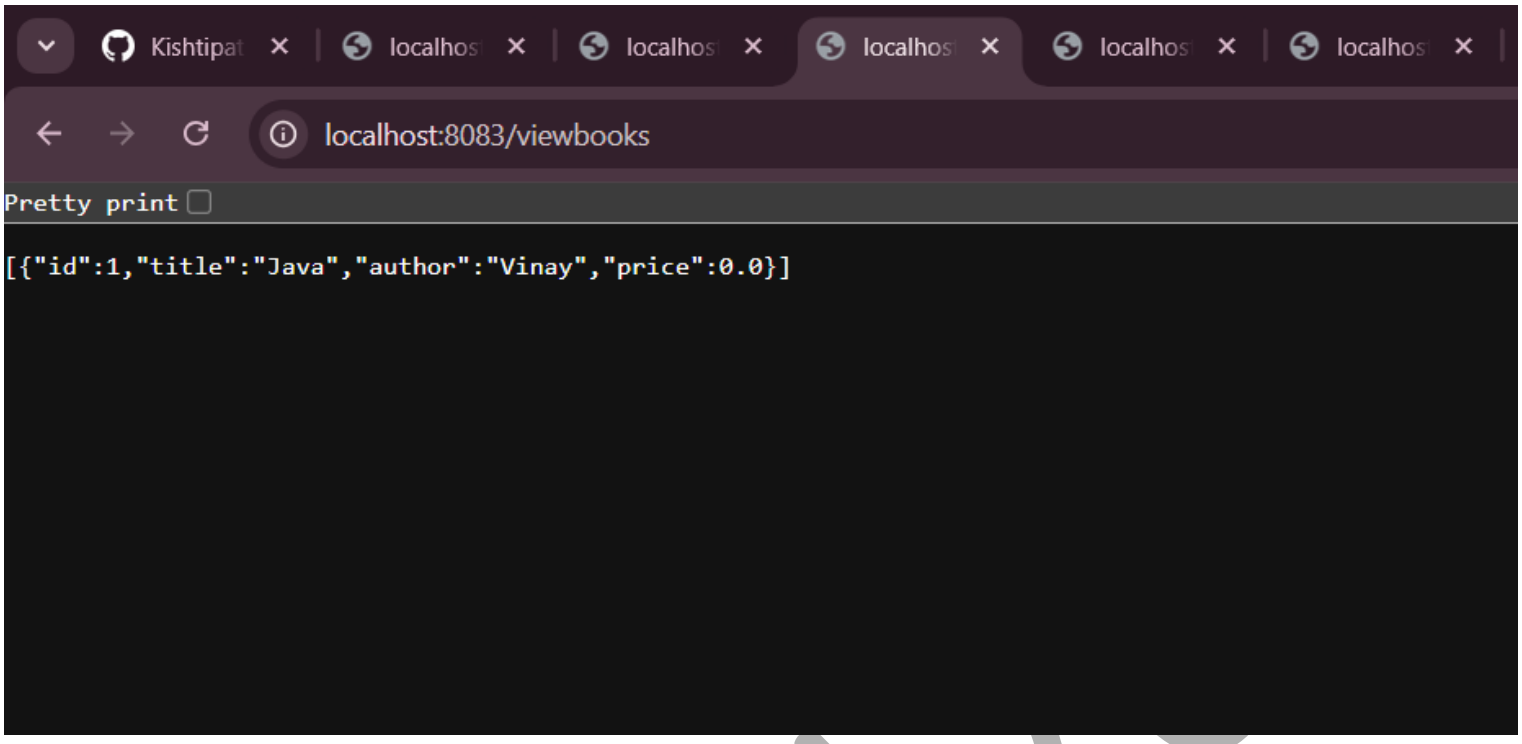
12-03-2026



GitHub Repository Link

The complete source code for this experiment is available in the following GitHub repository:

Repository:
https://github.com/Kishtipati-Vinay-Bhargav-Reddy/2400031280_FSAD_SKILL_6.git

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